Translation relations and a twist-free converse theorem at level 32

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Abstract

We prove the holomorphic Conrey–Farmer converse theorem at level 32 for every positive even weight and both Fricke eigenvalues. Only the Euler conditions at 2 and 3 are needed. The proof develops a signed two-term elliptic criterion, applies it uniformly to the rescaled $m = 3$ relations at levels $4, 8, 16, 32$, and completes the level 32 case by an explicit character of $\Gamma_0(8)$. The character is nontrivial on every cusp stabilizer, which proves cuspidality without additional coefficient bounds. We also record a parameterized version of the three-dimensional translation module used at level 36. This is a separate continuation of the level 36 work; that work is not modified here.

1 The new level and its hypotheses

The transformation-law results of Conrey and Farmer [1]; Conrey, Farmer, Odgers, and Snaith [2]; and Bedert, Cooper, Oliver, and Zhang [3] cover, collectively, $1 \le N \le 18$ and $N = 20, 23, 24$. Level 32 is outside these lists. The assertion below includes a separate proof of vanishing at all cusps; we do not infer cuspidality from the functional equation alone.

Theorem 1. *Let $k \ge 2$ be even and let $a_1 = 1$. Suppose*

$$F(s) = \sum_{n \ge 1} a_n n^{-s}$$

converges in some right half-plane and satisfies the following conditions:

1. *The function*
$$\Lambda(s) = \left(\frac{\sqrt{32}}{2\pi}\right)^s \Gamma(s) F(s)$$
extends to an entire function bounded in every closed vertical strip, and $\Lambda(s) = \omega \Lambda(k - s)$ for some $\omega \in \{1, -1\}$.

2. *$a_{2n} = 0$ for every $n \ge 1$.*

3. *For every $n \ge 1$,*
$$a_{3n} = a_3 a_n - 3^{k-1} a_{n/3}, \qquad a_{n/3} = 0 \quad \text{if } 3 \nmid n.$$

Then

$$f(z) = \sum_{n \ge 1} a_n e^{2\pi i n z} \ \in S_k(\Gamma_0(32)).$$

Both functional-equation signs are allowed, and there is no additional growth assumption on the coefficients.

Conditions (2)–(3) are precisely the coefficient conditions imposed by the partial Euler product

$$F(s) = \frac{1}{1 - a_3 3^{-s} + 3^{k-1-2s}} \sum_{(n,6)=1} a_n n^{-s}.$$

Consequently the full Conrey–Farmer Euler product at level 32 implies the local hypotheses of the theorem. No local factor at any prime other than 2 and 3 is used.

For $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$ use the normalized slash action

$$(v|_k M)(z) = (\det M)^{k/2}(cz + d)^{-k} v\left(\frac{az + b}{cz + d}\right).$$

We suppress the subscript. Since k is even, all nonzero scalar matrices act trivially. Put

$$T = \beta(1), \qquad \beta(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

$$H_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}, \qquad W_M = \begin{pmatrix} 1 & 0 \\ M & 1 \end{pmatrix}.$$

Convergence at a real point implies $a_n = O(n^A)$ for some $A \geq 0$. Thus f is holomorphic on $\mathfrak{H}$. Mellin inversion and the functional equation give

$$f|H_N = \varepsilon f, \qquad \varepsilon = (-1)^{k/2}\omega \in \{1, -1\}. \tag{1}$$

This is the usual Hecke argument, with the analytic root number ω distinguished from the slash eigenvalue ε.

2 Two generalizations of the level 36 method

2.1 The translation module with a parameter

Proposition 2. *Suppose h is holomorphic on $\mathfrak{H}$, $h|S = \varepsilon h$ with $\varepsilon^2 = 1$, and, for some $\tau \in \mathbb{C}$,*

$$h|(T^2 - \tau T + 1) = 0.$$

Set $g = h|T$ and $v = g|S$. The span of h, g, v is stable under $\mathrm{SL}_2(\mathbb{Z})$, with action

	$\|S$	$\|T$
h	εh	g
g	v	$\tau g - h$
v	g	$\tau h - \varepsilon v$

(2)

No linear independence is required. The formal matrices are

$$A_\varepsilon = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad B_{\tau,\varepsilon} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & \tau & 0 \\ \tau & 0 & -\varepsilon \end{pmatrix},$$

and they satisfy

$$A_\varepsilon^2 = (A_\varepsilon B_{\tau,\varepsilon})^3 = I, \qquad \det(XI - B_{\tau,\varepsilon}) = (X + \varepsilon)(X^2 - \tau X + 1).$$

Proof. The translation equation gives $h|T^{-1} = \tau h - g$. The identity $STS = T^{-1}ST^{-1}$, projectively, gives

$$\varepsilon v = h|STS = (\tau\varepsilon h - v)|T^{-1}.$$

Right multiplication by T yields $v|T = \tau h - \varepsilon v$. All the other entries in the table follow directly from the definitions. The matrix identities follow by multiplication. □

The values $\tau = -1, 0, 1$ correspond to Fourier frequencies with translation eigenvalues that are primitive third, fourth, and sixth roots, respectively. Thus the same construction applies after rescaling at levels $9, 16, 36$. The module itself does not establish cuspidality or the required congruence kernel in every case; these remain separate steps. In particular, the Eisenstein caveat at levels 9 and 16 in [1, Section 5] must be respected.

2.2 A signed two-term elliptic criterion

Lemma 3 (Elliptic vanishing)**.** *Let $k > 0$ be even and u holomorphic on $\mathfrak{H}$. If $u|E = \xi u$, where $E \in \mathrm{SL}_2(\mathbb{R})$ is elliptic of infinite order and ξ is a root of unity, then $u = 0$.*

Proof. Taking a positive power reduces to $u|E = u$, with E still elliptic of infinite order. If $a \in \mathfrak{H}$ is its fixed point, the coordinate $w = (z-a)/(z-\bar{a})$ identifies its action with $w \mapsto \zeta w$, where ζ has infinite order on the unit circle. Write $u(z)(dz)^{k/2} = U(w)(dw)^{k/2}$. Invariance gives

$$U(w) = \zeta^{k/2} U(\zeta w).$$

For $U(w) = \sum_{j\geq 0} c_j w^j$ it follows that $(1 - \zeta^{j+k/2})c_j = 0$. Since $j + k/2 > 0$, all c_j vanish. □

Lemma 4 (Symmetric pair)**.** *Suppose $h|T = \eta h$ with η a root of unity. Let $C \in \mathrm{SL}_2(\mathbb{R})$, $m \in \mathbb{Z}$, $t \in \mathbb{R}$, and $\lambda \in \{1, -1\}$. Put $C' = T^m C^{-1} T^{-m}$. If*

$$h|(1 - \lambda C)\beta(-t) + h|(1 - \lambda C')\beta(t) = 0 \tag{3}$$

and

$$E = C^{-1}\beta(2t - m)$$

is elliptic of infinite order, then $h|C = \lambda h$.

Proof. Let $u = h|(1 - \lambda C)$. Using only the known scalar action of T^m, we obtain

$$h|(1 - \lambda C') = \eta^m h|(1 - \lambda C^{-1})T^{-m} = -\lambda\eta^m u|C^{-1}T^{-m}.$$

Thus (3), right-multiplied by $\beta(t)$, becomes

$$u = \lambda\eta^m u|C^{-1}\beta(2t - m).$$

Apply Lemma 3. □

This formulation permits signed transformation laws and integer translation conjugacies. It is useful to check the trace directly: if $C = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, then

$$\mathrm{tr}\big(C^{-1}\beta(s)\big) = a + d - cs.$$

A determinant-one real matrix with rational, nonintegral trace in $(-2, 2)$ is elliptic of infinite order. Indeed, finite order would make its trace a rational algebraic integer, hence an integer.

3 The uniform rescaled relation at levels $4, 8, 16, 32$

We first record the needed consequence of a good Euler factor, including the justification of the left replacements in the group algebra.

Lemma 5 (Prime relation)**.** *Suppose $f|T = f$, $f|H_N = \varepsilon f$, $\varepsilon^2 = 1$, and for a prime $p \nmid N$ the coefficients of f satisfy $a_{pn} = a_p a_n - p^{k-1} a_{n/p}$. For each nonzero residue class b modulo p, choose an integer c such that $Nbc \equiv -1 \pmod{p}$, and put*

$$\gamma_b = \begin{pmatrix} p & -b \\ -Nc & (1+Nbc)/p \end{pmatrix}.$$

Then

$$f\Big| \sum_{b \bmod p}^{*} (1-\gamma_b)\beta(b/p) = 0. \tag{4}$$

Proof. The coefficient recurrence is equivalent to

$$f|\mathcal{T}_p = p^{1-k/2} a_p f, \qquad \mathcal{T}_p = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} + \sum_{r=0}^{p-1} \begin{pmatrix} 1 & r \\ 0 & p \end{pmatrix}.$$

Conjugate by H_N and subtract. The two diagonal terms cancel. Right multiplication by $\mathrm{diag}(p, 1)$ gives

$$f\Big| \sum_{r=1}^{p-1} \beta(r/p) = f\Big| \sum_{c=1}^{p-1} \begin{pmatrix} p^2 & 0 \\ -Ncp & 1 \end{pmatrix}.$$

Periodicity permits any representatives for r. Also $f|W_N = f$, since $W_N = H_N T^{-1} H_N^{-1}$. Replacing c by $c + pj$ on the right is therefore permitted: it amounts to left multiplication by W_N^{-j}. Reindex by $Nbc \equiv -1 \pmod{p}$ and use the exact identity

$$\gamma_b \beta(b/p) = \begin{pmatrix} p & 0 \\ -Nc & 1/p \end{pmatrix}.$$

Scalar matrices do not affect the slash action, giving (4). This is the prime case of [1, Corollary 2]. □

Now let $N = 4M$ with $3 \nmid M$. Suppose f has the functional equation at level N, $a_{2n} = 0$, and the good Euler recurrence at 3. Put $D = \mathrm{diag}(1, 2)$ and

$$h(z) = f(z/2) = 2^{k/2}(f|D)(z).$$

Odd Fourier support and Fricke conjugation give

$$h|T = -h, \qquad h|H_M = \varepsilon h, \qquad h|W_M = -h. \tag{5}$$

The last equality uses $W_M = H_M T^{-1} H_M^{-1}$ and $\varepsilon^2 = 1$.

Define $\sigma \in \{1, -1\}$ by $\sigma M \equiv -1 \pmod{3}$. In Lemma 5, choose $b = -1, c = -\sigma$ and $b = 1, c = \sigma$. After conjugating by D, the relation is

$$h|(1-A)\beta(-2/3) + h|(1-B)\beta(2/3) = 0, \tag{6}$$

where

$$A = \begin{pmatrix} 3 & 2 \\ 2\sigma M & (1+4\sigma M)/3 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & -2 \\ -2\sigma M & (1+4\sigma M)/3 \end{pmatrix}.$$

Set

$$C = -W_M^{-\sigma} A = \begin{pmatrix} -3 & -2 \\ \sigma M & (2\sigma M - 1)/3 \end{pmatrix}, \qquad C' = -W_M^{\sigma} B = \begin{pmatrix} -3 & 2 \\ -\sigma M & (2\sigma M - 1)/3 \end{pmatrix}. \tag{7}$$

All four matrices have determinant one. Because $h|W_M^{\pm 1} = -h$, (6) becomes

$$h|(1+C)\beta(-2/3) + h|(1+C')\beta(2/3) = 0. \tag{8}$$

A direct multiplication gives

$$C' = \beta(r) C^{-1} \beta(-r), \qquad r = \frac{2}{3} + \frac{8}{3\sigma M}. \tag{9}$$

Proposition 6. *Under the assumptions just made, if $M \in \{1, 2, 4, 8\}$, then $h|C = -h$ for the matrix C in* (7).

Proof. For these four values, r is an integer. Apply Lemma 4 with $\eta = -1$, $\lambda = -1$, $m = r$, and $t = 2/3$. The determinant-one matrix

$$E = C^{-1}\beta(4/3 - r)$$

has trace

$$\frac{2\sigma M - 10}{3} - \sigma M \left(\frac{2}{3} - \frac{8}{3\sigma M} \right) = -\frac{2}{3}.$$

It is therefore elliptic of infinite order, proving the assertion. □

M	$N = 4M$	σ	r	$\operatorname{tr} E$
1	4	-1	-2	$-2/3$
2	8	1	2	$-2/3$
4	16	-1	0	$-2/3$
8	32	1	1	$-2/3$

The range is exact for this particular integer-translation argument: if r in (9) is integral, then $\sigma M(3r-2) = 8$, so M divides 8. For other M, the same matrix identities remain valid, but (5) does not provide a scalar action of the nonintegral translation $\beta(r)$. No such scalar action is assumed here.

4 Completion of the level 32 transformation law

Specialize to $M = 8$. We have

$$A = \begin{pmatrix} 3 & 2 \\ 16 & 11 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & -2 \\ -16 & 11 \end{pmatrix}, \qquad C = \begin{pmatrix} -3 & -2 \\ 8 & 5 \end{pmatrix}, \qquad C' = \begin{pmatrix} -3 & 2 \\ -8 & 5 \end{pmatrix}.$$

For clarity, the complete defect calculation in this case is

$$W_8^{-1} A = -C, \qquad W_8 B = -C', \qquad C' = T C^{-1} T^{-1}.$$

Let $u = h|(1+C)$. Since $h|T = -h$,

$$h|(1+C') = -u|C^{-1}T^{-1}.$$

Equation (8) gives

$$u = u|E, \qquad E = C^{-1}\beta(1/3) = \begin{pmatrix} 5 & 11/3 \\ -8 & -17/3 \end{pmatrix}, \qquad \operatorname{tr} E = -2/3. \tag{10}$$

Thus $u = 0$ and

$$h|T = h|W_8 = h|C = -h. \tag{11}$$

Lemma 7. *Projectively,*

$$\Gamma_0(8) = \langle T, W_8, C\rangle.$$

Proof. First, projectively, $\Gamma(2)$ is generated by T^2 and $\left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$. Here is the elementary reduction argument. In the first column of an element of $\Gamma(2)$, the top entry is odd and the bottom entry is even. Left multiplication by powers of these two generators subtracts even multiples of one entry from the other. Choosing a nearest multiple makes the new absolute value strictly smaller than the other one: equality is impossible because of the different parities. Repeating reduces the bottom entry to zero, after which the matrix is a power of T^2, up to sign.

Conjugation by $D = \operatorname{diag}(1,2)$ identifies $\Gamma(2)$ with $\Gamma_0(4)$ and carries these generators to T, W_4. The homomorphism

$$\Gamma_0(4) \longrightarrow \mathbb{Z}/2\mathbb{Z}, \qquad \begin{pmatrix} a & b \\ 4c & d \end{pmatrix} \longmapsto c \pmod 2$$

has kernel $\Gamma_0(8)$. Using representatives $1, W_4$, Schreier's lemma gives generators

$$T, \qquad W_4^2 = W_8, \qquad W_4 T W_4^{-1}.$$

Finally,

$$C = (W_4^{-1}T)^2, \qquad W_4 T W_4^{-1} = W_8 C T^{-1}.$$

This proves the assertion, and also verifies directly the generating set displayed in [1, Section 2]. □

For $\gamma = \left(\begin{smallmatrix} a & b \\ 8c & d \end{smallmatrix}\right) \in \Gamma_0(8)$ define

$$\chi(\gamma) = (-1)^{b+c}. \tag{12}$$

This is a character. Indeed a, d are odd, and multiplication adds both b and c modulo 2. It is trivial on $-I$ and takes the value -1 on each of T, W_8, C. Thus (11) and Lemma 7 imply

$$h|\gamma = \chi(\gamma)h \qquad (\gamma \in \Gamma_0(8)). \tag{13}$$

For every $\gamma = \left(\begin{smallmatrix} a & b \\ 32c & d \end{smallmatrix}\right) \in \Gamma_0(32)$ we have

$$D^{-1}\gamma D = \begin{pmatrix} a & 2b \\ 16c & d \end{pmatrix} \in \Gamma_0(8), \qquad \chi(D^{-1}\gamma D) = (-1)^{2b+2c} = 1.$$

Since $h = 2^{k/2} f|D$, equation (13) proves $f|\gamma = f$. Only inclusion of the conjugated group in $\ker\chi$ is needed; these two groups are not asserted to be equal.

5 Vanishing at every cusp

The coefficient bound already obtained implies

$$f(x+iy) = O(y^{-A-1}) \qquad (0 < y \le 1),$$

uniformly in x. It follows that $h|\alpha$ has at most polynomial growth as $y \to \infty$, uniformly for x in a bounded interval, for every $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. To see this when the lower-left entry of α is nonzero, use $\mathrm{Im}(\alpha z) = \det(\alpha) y/|cz+d|^2 \asymp y^{-1}$ and the factor $(cz+d)^{-k}$. When the lower-left entry is zero, the Fourier series instead gives exponential decay.

The cusps of $\Gamma_0(8)$ are represented by $\infty, 0, 1/2, 1/4$, with respective widths $1, 8, 2, 1$. For a direct completeness check, the four denominators have distinct greatest common divisors with 8, so the cusps are inequivalent; their widths sum to $12 = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(8)]$. They therefore exhaust the orbits of the parabolic action on the twelve cosets. Representatives and their primitive stabilizers are:

Cusp	Width w	$P = \alpha T^w \alpha^{-1}$	$\chi(P)$
∞	1	T	-1
0	8	W_8^{-1}	-1
1/2	2	$\begin{pmatrix} -3 & 2 \\ -8 & 5 \end{pmatrix}$	-1
1/4	1	$\begin{pmatrix} -3 & 1 \\ -16 & 5 \end{pmatrix}$	-1

Here take $\alpha = I, S, \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 & 0 \\ 4 & 1 \end{smallmatrix}\right)$, respectively. Equation (13) shows that

$$(h|\alpha)(z+w) = -(h|\alpha)(z).$$

Consequently $h|\alpha$ has a Fourier expansion of period $2w$ containing only odd indices. Polynomial growth excludes every negative index: equivalently, the Laurent series in $q = e^{2\pi i z/(2w)}$ has no negative powers, by Cauchy's coefficient estimate and the bound by a power of $\log(1/|q|)$. Anti-periodicity excludes the constant term. Thus $h|\alpha$ decays exponentially at every one of these cusps.

Every rational cusp is equivalent to one of them under $\Gamma_0(8)$. The character law therefore gives vanishing at every rational cusp, also when h is viewed on the subgroup $D^{-1}\Gamma_0(32)D$. Rational rescaling preserves this vanishing. Hence f vanishes at every cusp of $\Gamma_0(32)$, completing the proof of Theorem 1.

6 What has and has not been generalized

The translation-module calculation now has the arbitrary parameter τ of Proposition 2. The defect argument is the signed criterion of Lemma 4, which is independent of the level. The half-translation reduction works for every N divisible by 4, and its $m = 3$ matrix identities work whenever $3 \nmid N$. The integer-conjugacy step closes uniformly for $N = 4, 8, 16, 32$. Among these, level 32 supplies the additional converse theorem.

This argument does not establish the converse theorem at arbitrary $4M$. For example, if $M = 7$, the parameter in (9) is $2/7$, and the known anti-periodicity under T does not control $\beta(2/7)$. The preceding proofs do not replace it by a scalar. Further levels require another relation or a different factorization.

The accompanying program `verify_N32.py` checks the rational matrix identities, the parameterized three-dimensional identities, the signed characters, the four cusp orbits and widths, and the elliptic trace. It also verifies the recorded SHA-256 hashes of the three original level 36 deliverables when they are present next to this new directory. The mathematical proofs of generation, elliptic vanishing, and cusp decay are given above and do not depend on a finite numerical search.

References

[1] J. B. Conrey and D. W. Farmer, *An extension of Hecke's converse theorem*, International Mathematics Research Notices 1995, no. 9, 445–463. arXiv:math/9502209.

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